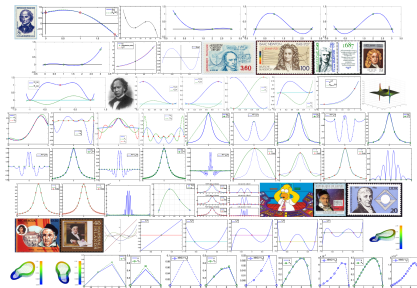


MA378 Chapter 5: Review

§5.1 Review (and Exam Preview)

Dr Niall Madden (Niall.Madden@UniversityOfGalway.ie)

Start: 1 March 2026 (W12.1)



1.0 Review/Preview:

1 Review

- Check List

- §1: Polynomial Interpolation

- §2: Piecewise Polynomial Interpolation

- §3 Numerical Integration (“Quadrature”)

2 §4: Finite Element Methods

3 MATLAB

- Context

4 Assessment

- The Exam

5 Thanks

These notes are posted only to Canvas, and not to the usual location of <https://www.niallmadden.ie/2526-MA378/>

This module has four chapters:

1. Polynomial Interpolation
2. Piecewise polynomial interpolation
3. Numerical Integration (Quadrature)
4. Finite element methods

We also studied implementation of methods in MATLAB.

We now review these — primarily to provide you with a check-list of topics.

1.1 The polynomial interpolation problem (PIP) in 2 forms;

1.2 Uniqueness (if $p_n \in \mathcal{P}_n$ has $n + 1$ zeros, then $p_n \equiv 0$)

1.2 ~~The Vandermonde matrix method (and its ill-conditioning)~~

1.2 **Lagrange (Basis) Polynomials**

1.2 **Lagrange Interpolation**

1.3 Rolle's Theorem

1.3 **Cauchy's theorem:** $f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!} \pi_{n+1}(x).$

1.4 Hermite Interpolation: construction and error estimates.

1.5 Convergence & Runge's Example

2.1 Linear Interpolating Splines:

- ▶ construction,
- ▶ error analysis,
- ▶ best approximation, and
- ▶ the Minimum Energy property.

2.2 (Natural) Cubic Splines:

- ▶ definition
- ▶ construction
- ▶ Error analysis (without proof)

2.3 PCHIP: Piecewise Cubic Hermite Interpolating Polynomial.

- ▶ definition
- ▶ construction,
- ▶ error analysis.

- 3.1 Newton-Cotes methods, and the Trapezium Rule
- 3.1 Undetermined Coefficients
- 3.2 Simpson's Rule and (non-sharp) error estimates
- 3.3 Precision
- 3.3 Composite rules
- 3.4 Gaussian Quadrature (and Undetermined Coefficients again)
- 3.5 A sequence of Orthogonal Monic Polynomials (and vector spaces and inner products); properties of orthogonal monic polynomials; their roots as quadrature points.
- 3.6 Precision and convergence of Gaussian Quadrature methods.

1.2 §4: Finite Element Methods

- 4.1 Boundary value problems, and Maximum Principles.
- 4.1 The **variational** (weak) **formulation** of the problem, and the uniqueness of its solution.
- 4.2 The **FEM**, including finite and infinite dimensional spaces.
- 4.2 **FE implementation**, including Galerkin basis functions (“hat functions”), how to construct the linear system, and why the system matrix is tridiagonal and symmetric.
- 4.3 Analysis, **Cea’s Lemma**.

1.3 MATLAB

We also studied some of these methods using MATLAB. There were several reasons for this:

- ▶ It is a language/system designed by/for numerical analysts;
- ▶ It is new to all of you;
- ▶ We could study methods experimentally.
- ▶ Although it appears niche, it is used in industry. Locally, that includes Medtronic, Boston Scientific, Stryker and Valeo. Nationally: Analog Devices, Central Bank of Ireland, EirGrid, Advanced Micro Devices (AMD), Qualcomm, Johnson & Johnson, Pfizer.

Numerical Analysis is an ancient field of mathematics.

The origins of the topics we've studied have roots in astronomy: **interpolation** techniques were used in astronomy. Earliest known examples date from use of linear interpolation in 200 BC (Babylon; modern Iraq), polynomial interpolation in 600 AD (China), and 650 AD (India).

Quadrature (numerical integration) has been studied for about as long – and long before the invention/discovery of calculus. The Trapezium Rule was known to the Babylonians, and was used by the ancient Greeks for tasks such as estimating π .

Finite Elements have a more recent history, and have been developed since the 1950, starting with the work of Richard Courant.

All three area feature in modern mathematical research.

- ▶ Since about 2010 there has been renewed interest in robust methods for very high-order polynomial interpolation, and Chebychev methods in particular;
- ▶ Modern research in quadrature focuses on fast methods for high-dimensional problems;
- ▶ 1,000s of papers are published every year on finite element methods, with much effort aimed at understanding how to construct meshes automatically, use of discontinuous basis functions, and applications to non-linear problems.

So: in spite of the long history, what you have studied is very relevant to modern mathematics.

1.4 Assessment

MA378 is assessed by

- ▶ Labs (10%). Lab 3 still to be graded.
- ▶ Two Assignments (10%). (Assignment 2 still not graded; sorry!)
- ▶ Class test (10%)
- ▶ Written two-hour exam (70%)

The exam will have **4 questions**: answer all 4.

The layout is different to recent years.

- ▶ There are 4 questions, all worth 25 marks.
- ▶ Q1 is on Section 1 (Polynomial interpolation)
- ▶ Q2 is on Section 2 (Piecewise polynomial interpolation)
- ▶ Q3 features material from Section 3 (quadrature)
- ▶ Q4 features material from Sections 3 and 4 (BVPs + FEMs).

For the most part, the emphasis is on derivation of methods, and demonstration of understanding, and use of analysis. But it does feature some proofs.

MATLAB will not feature on the exam.

Important: the above guide is only for the **Semester 2 2025/2026** exam. If there is an Autumn exam, the layout could be different again.

Tips (this exam only)

- ▶ The formula for the error in Polynomial Interpolation will be given on the paper.
- ▶ Unusually, you will not be asked to prove Cauchy's Theorem.
- ▶ Lagrange (basis) polynomials are very important: be able to define these, and give a formula.
- ▶ Topics that have not appeared on recent papers, could feature, such as Hermite Interpolation/PCHIPs.
- ▶ Make sure you understand the concept of **precision** for Quadrature rules.
- ▶ Be ready for the type of questions that ask “what value of N is needed to ensure...”
- ▶ I do like a good uniqueness proof.

1.5 Thanks

Thanks!

Thanks for taking this module. I hope you have found it interesting, and that some of the topics covered are useful in your future mathematical and computational efforts.

Feedback from graduates suggests its one of the most useful “MA” modules. If you are interested in graduate studies in this field, in Ireland or elsewhere, please talk to me.

1.5 Thanks

